Auditing Model Substitution in LLM APIs
Why Software Tests Fail and Hardware Attestation Wins

Are You Getting What You Pay For?

Based on arXiv:2504.04715

The Model Substitution Problem: Economic Incentives Drive Covert Swaps

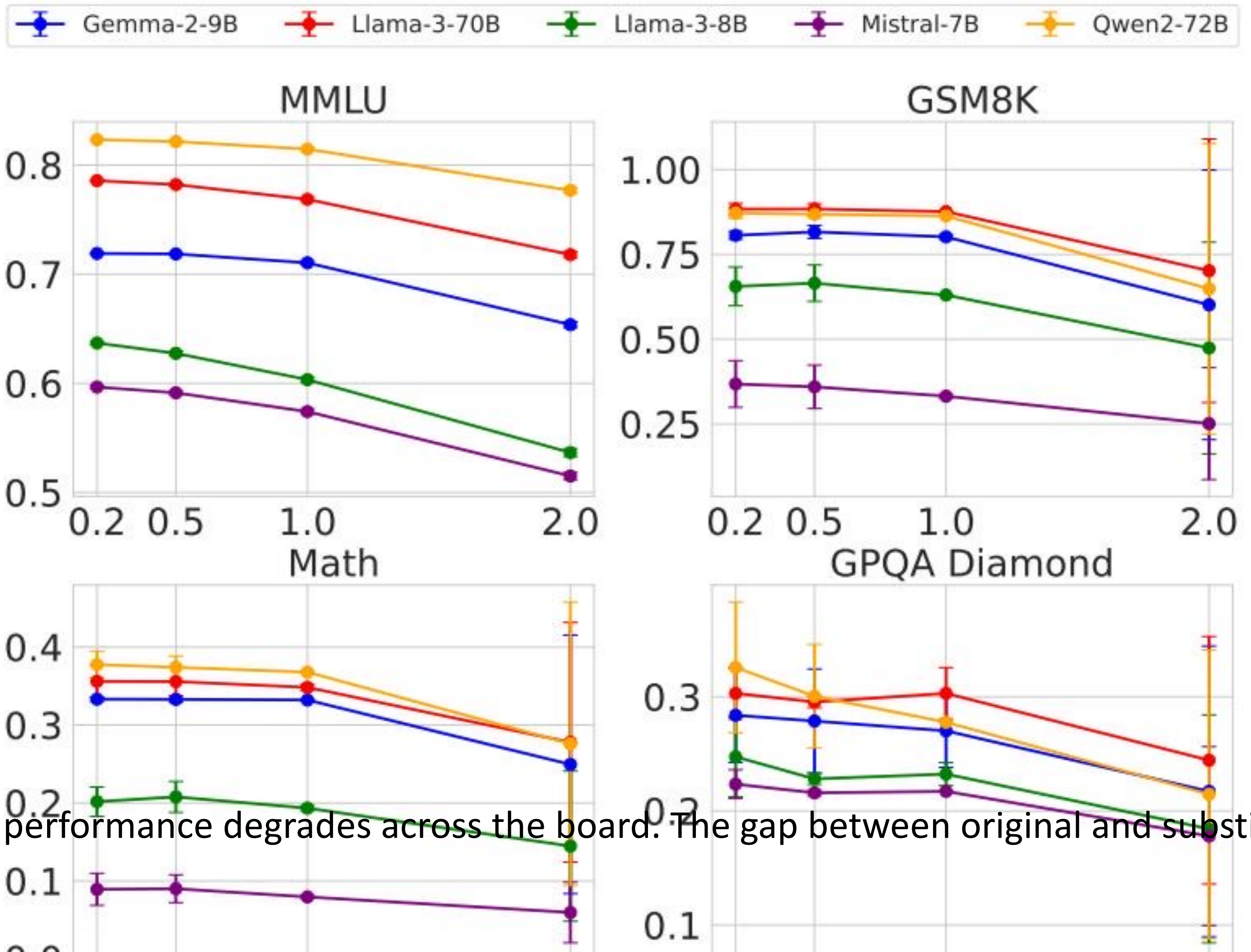
- Providers face economic incentives to substitute cheaper models to cut GPU costs.
- Four main types of substitution:
  - 1. Smaller Models (e.g., 8B instead of 70B)
  - 2. Quantized Variants (e.g., FP8/INT8 instead of FP16)
  - 3. Fine-tuned or Updated Versions
  - 4. Entirely Different Model Families
- Formalized as Hypothesis Testing: Can an auditor distinguish $P\_spec(y|x)$ from

FP8 Quantization Preserves Most Accuracy — Making Subst

Model	MMLU	GSM8K	MATH	GPQA Diamond
Llama-3-8B-Instruct	62.69 ± 0.18	61.14 ± 3.47	20.65 ± 3.43	22.62 ± 0.22
Llama-3-8B-Instruct-FP8	62.43 ± 0.26	60.90 ± 4.10	14.91 ± 2.49	20.14 ± 0.24
Llama-3-70B-Instruct	78.05 ± 0.08	88.06 ± 1.44	35.69 ± 1.33	29.60 ± 0.51
Llama-3-70B-Instruct-FP8	77.88 ± 0.13	87.35 ± 1.24	35.75 ± 1.16	33.12 ± 0.30
Gemma-2-9b-it	71.86 ± 0.08	81.80 ± 1.35	33.41 ± 0.28	28.64 ± 2.97
Gemma-2-9b-it-FP8	71.92 ± 0.11	79.41 ± 1.14	32.53 ± 0.34	27.81 ± 3.22
Qwen2-72B-Instruct	82.18 ± 0.08	86.72 ± 1.00	37.39 ± 1.41	29.93 ± 2.71
Qwen2-72B-Instruct-FP8	81.98 ± 0.08	86.82 ± 0.97	37.67 ± 1.39	31.08 ± 1.96
Mistral-7B-Instruct-v0.3	59.15 ± 0.10	35.90 ± 4.54	8.94 ± 1.24	21.60 ± 0.17
Mistral-7B-Instruct-v0.3-FP8	58.77 ± 0.13	32.20 ± 4.02	7.68 ± 1.23	22.72 ± 0.19

Performance drops are often within error bars, complicating statistical detection.

Higher Temperature Amplifies Performance Gaps but Increas



At temperature 2.0, performance degrades across the board. The gap between original and substitute narrows or becomes no

Log Probability Fingerprinting Fails Due to Inference Nondet

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20

Llama-3-8B produces varying log probabilities across software versions (transformers, vLLM) and hardware (A100, H100). This

Randomized Routing Makes Detection Query-Prohibitive

- Randomized Substitution Attack:
  - Provider mixes substitute responses with probability p.
  - Mixed distribution: $P_{mixed} = (1-p) * P_{spec} + p * P_{substitute}$
  - As p approaches 0, P_{mixed} converges to P_{spec}.
  - Requires exponentially more queries to detect statistically.
- Combined with Benchmark Evasion:
  - Providers route known benchmark prompts to genuine models.

Software-Only Auditing Methods Are Fundamentally Unreliab

Method Type	Weakness	Practical Limitation
Text-Based Statistical Tests (e.g., MMD)	Fail against subtle substitutions (e.g., FP8)	Require impractically large query budgets
Log-Probability Fingerprinting	Defeated by production-level nondeterminism	False positives from different hardware/software (vLLM, A100 vs H100)
Adversarial Countermeasures	Benchmark evasion & randomized routing (low p)	Fundamentally query-prohibitive and easily bypassed

Trusted Execution Environments (TEEs) Provide Provable Integrity

- Hardware-Secured Enclaves:
  - Cryptographic attestation of the code and data being executed.
  - Guarantees that specific model weights are loaded.
- Provable, Efficient, Robust:
  - Modest performance overhead compared to computationally prohibitive ZKPs.
  - Immune to software-level evasion techniques.
- Practical Deployment Path:

Key Takeaways: Close the Verification Gap with Hardware Attestation

- 1. Model substitution is economically motivated and practically undetectable via software.
- 2. FP8 quantization barely affects benchmarks, masking substitution.
- 3. Inference nondeterminism kills log-prob fingerprinting.
- 4. Randomized substitution & evasion defeat statistical tests.
- 5. TEEs are the only currently viable path to